COM3031-Probabilistic Machine Learning Workshop: Variational Inferenc

Overview

In this workshop, we study how to minimize the Kullback–Leibler (KL) divergence between a target distribution p and a parametric distribution q by optimizing the parameters of q. We use the Jupyter notebook provided on ELE to compare forward vs. reverse KL objectives and to investigate their behaviour. We also connect reverse KL to Variational Inference (VI) via the Evidence Lower Bound (ELBO) in a separate notebook. Reviewing the Week 6 slides/notes will be helpful.

Questions

Question 1. Minimizing KL. Use the Jupyter notebook provided on ELE for all parts of this question.

(a) Compare forward KL and reverse KL as optimization objectives.

(b) Investigate the effect of different numbers of optimization iterations on the KL minimization.

(c) Investigate the effect of different initial parameter values of q on the KL minimization.

Question 2. VI via ELBO (Reverse KL). This question uses VI: optimize $q(z)$ by maximizing the ELBO (i.e., minimizing reverse KL $D_{\mathrm{KL}}(q(z) \,\|\, p(z \mid x))$. Please refer to the Jupyter notebook provided on ELE and complete the required steps.